

Program : Diploma in Civil and Environmental Engineering	
Course Code: 6352A	Course Title: Industrial Pollution and its Control
Semester: 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide knowledge about the effects of emissions from industries and control methods
- To equip the students with the knowledge about various air pollution control methods
- To facilitate the students with basic concepts of solid waste management.

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain the adverse effects of emissions from Industries and understand the guidelines set by the environmental protection agencies.	14	Understanding
CO2	Describe air pollution control methods and equipment.	14	Understanding
CO3	Select the appropriate treatment method required for treating waste water.	16	Understanding
CO4	Explain different techniques for municipal solid waste and hazardous waste management.	14	Understanding
	Series test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2					3		
CO3					3		
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the adverse effects of emissions from Industries and understand the guidelines set by the environmental protection agencies.		
M1.01	Discuss the different sources of pollution from industries.	2	Understanding
M1.02	Discuss the Environmental regulatory legislations and standards.	2	Understanding
M1.03	Explain the effects of pollution on living and non living things.	2	Understanding
M1.04	Describe the concept of Sustainable Development	2	Understanding
M1.05	Explain pollutant sampling and measurement	3	Understanding
M1.06	Describe the pollution control in industries.	3	Understanding

Contents

Industrial pollution: Different types of wastes generated in an industry, Different water pollutants, Air pollutants and solid wastes from industry, Their effects on living and non-living things, Environmental regulatory legislations and standards, Importance of industrial pollution abatement, Concept of sustainable development, Greenhouse gases, Global warming and climate change.

Pollutant sampling and measurement: Ambient air sampling: collection of gaseous air pollutants, collection of particulate air pollutants. Stack sampling: Sampling system, particulate sampling, and gaseous sampling.

Pollution Control in Industries: Manufacturing process, Identification & characterization of sources of wastewater, treatment of wastewater including recycling & reuse concepts in

Textile industry, pharmaceutical industry, Diary industry, sugar industry, starch industry, fertilize industry, tannery, distillery, pulp & paper industry, petrochemical industry, dye & dye intermediate.

CO2	Describe air pollution control methods and equipment.		
M2.01	Explain the analysis of air pollutants	4	Understanding
M2.02	Understand the general methods of control and removal of gaseous pollutants	3	Understanding
M2.02	Explain the workings of particulate control equipment like gravitational settling chambers, cyclone separators etc.	4	Understanding
M2.03	Describe the methods to be adopted for the control of gaseous emissions	3	Understanding

Contents

Analysis of air pollutants: Sulphur dioxide, nitrogen oxides, carbon monoxide, oxidants and Ozone, hydrocarbons, particulate matter. General methods of control and removal of sulfur dioxide, oxides of nitrogen and organic vapors from gaseous effluent

Air pollution control methods and equipment: Source collection methods: raw material changes, process changes, and equipment modification. Cleaning of gaseous effluents particulate emission control: collection efficiency, particulate control equipment like gravitational settling chambers, Cyclone separators, fabric filters, ESP and their constructional details and design aspects. Scrubbers: wet scrubbers, spray towers, centrifugal scrubbers, packed beds and plate columns, venturi scrubbers, their design aspects.

Control of gaseous emissions: absorption by liquids, absorption equipment, adsorption by solids, equipment and the design aspects.

CO3	Select the appropriate treatment method required for treating waste water.		
M3.01	Explain the characteristics of sewage and the significance of tests on sewage	3	Understanding
M3.02	Describe the methods of primary treatment of waste water	3	Understanding
M3.03	Outline the functions and working of screens, skimming tanks, and grit chambers.	3	Understanding
M3.04	Describe various sewage treatment stages	4	Understanding
M3.05	Explain the different disposal methods	3	Understanding

Contents:

Sewage characteristics: - Strength of sewage, sampling of sewage, characteristics of sewage - physical, chemical and biological - significance of the following tests for -Solids, Oxygen demand, BOD, COD, pH value, Chlorides

Introduction to wastewater treatment. Preliminary treatment : brief description and functions of Screens, Skimming tanks and Grit chambers.

Primary treatment - brief description and functions of plain sedimentation

Biological treatment of wastewater, bacterial and bacterial growth curve, aerobic processes, suspended growth processes, activated aerated lagoons and stabilization ponds, Attached growth processes, trickling filters, rotary drum filters, anaerobic processes.

Methods of tertiary treatment: A brief study of carbon absorption, ion exchange, reverse osmosis, ultrafiltration, chlorination, ozonation, treatment and disposal.

Sewage disposal - dilution, disposal on lands, oxidation ponds, oxidation ditch, aerated lagoons, anaerobic lagoons. (brief description only)

CO4	Explain different techniques for municipal solid waste and hazardous waste management.		
M4.01	Explain the sources and health and environmental effects of nuclear waste.	2	Understanding
M4.02	Describe the sources and disposal methods of nuclear wastes.	3	Understanding
M4.03	Outline the treatment and disposal methods of chemical wastes	3	Understanding
M4.04	Explain the types of biomedical waste and their control	3	Understanding
M4.05	Outline the safety considerations in industries	3	Understanding

Contents:

Municipal solid waste sources, and control methods. Hazardous waste management- Nuclear wastes: health and environment effects, sources and disposal methods.

Chemical wastes: health and environmental effects, treatment and disposal: treatment and disposal by industry, off site treatment and disposal, treatment practices in various countries.

Biomedical wastes: types of wastes and their control.

Industrial hazards and Safety considerations in chemical industries- Chemical, Electrical and mechanical hazards, Fire and explosion hazard, Health hazard, Laboratory Safety, Safety Practices, Factory acts.

Text / Reference:

T/R	Book Title/Author
T1	Rao. C.S., “Environmental Pollution and Control Engineering”, 2nd Edition, Revised, New Age International, 2007
T2	Mahajan. S.P., “Pollution Control in Process Industries”, Tata-McGraw Hill, New Delhi, 1985.
R3	M. Narayana Rao and A.K. Datta , “Wastewater Treatment” ,Oxford and IHB publ. New Delhi.
R4	Swamy, A.V.N., “Industrial Pollution Control and Engineering”, Galgotia Publications, Hyderabad, 2005.
R5	B. K. Sharma, “Environmental Chemistry”, Krishna Prakashan Media (P) Ltd., Meerut.
R6	Metcalf & Eddy, “Wastewater Engineering”, Tata McGraw-Hill Education Private Limited (2009).
R7	Wark, K., Warner, C.F., and Davis, W.T., “Air Pollution: Its Origin and Control”, Addison-Wesley Longman. 1998.

Online resources:

Sl.No.	Website Link
1	https://onlinecourses.nptel.ac.in/noc23_ce14/preview
2	https://archive.nptel.ac.in/courses/103/107/103107215/
3	https://archive.nptel.ac.in/courses/105/107/105107213/
4	https://nptel.ac.in/courses/105107207